

Introducción AWS Glue DataBrew



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ENERO 23 Y 24
ALL DAY

Agenda



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23 Y 24 DE ENERO, 2021

- Presentación
- Qué es Glue Data Brew
- Conceptos previos
- Demostración

Presentación



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- Universidad Católica de Santa María - Ingeniería de Sistemas
- AWS Certified Solutions Architect - Associate
- Data Engineer - Morris & Opazo - Argo Cloud Solutions (Perú)
- Embajadora Women Techmakers y Organizadora GDG Arequipa
- Facilitadora of #IamRemarkable



¿Qué es Glue DataBrew?



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AWS Glue DataBrew es una herramienta de preparación de datos visual que permite a los usuarios limpiar y normalizar datos sin escribir ningún código.

- Descubrir
- Visualizar
- Limpiar
- Transformar



Glue DataBrew

Project (Proyecto) El espacio de trabajo de preparación de datos.

Projects (5) Info				
Find projects				
☐	Project name ▾	Associated dataset ▾	Attached recipe ▾	Jobs ▾
☐	cliente-tarjetav2	TipoDeTarjeta-Cliente	cliente-tarjetav2-recipe	clientes-tarjetas
☐	cliente-tarjeta	TipoDeTarjeta-Clientev1	cliente-tarjeta-recipe	-
☐	dbventas-order	dbventas-salesorders	dbventas-order-recipe	order
☐	dbventas-clientes	dbventas-salescustomers	dbventas-clientes-recipe	customer
☐	demo1	MisCanciones	demo1-recipe	-

DataSet (Conjunto de datos)

☐	Dataset name ▾	Data type ▾	Data profile ▾	Source ▾
☐	TipoDeTarjeta-Cliente	CSV	-	S3
☐	TipoDeTarjeta-Clientev1	CSV	-	S3
☐	dbventas-salesorders	Data Catalog table	-	Data Catalog
☐	archivoTipoDeTarjetaCliente	Data Catalog table	-	Data Catalog
☐	dbventas-salescustomers	Data Catalog table	-	Data Catalog
☐	MisCanciones	CSV	-	S3

↑ File upload
Data lake/data store
🗑 Amazon S3
AWS Glue Data Catalog
🗑 Amazon S3 tables
📄 Amazon Redshift tables
📄 Amazon RDS tables
🔗 All AWS Glue tables
Others
🔄 AWS Data Exchange

Recipe (Recipe): Conjunto de instrucciones o pasos a realizar sobre un dataset.

Applied steps (26) Clear all  
1. Split column on a single delimiter ; in <code>customer_id;customer_card</code>
2. Rename <code>customer_id;customer_card_1</code> to <code>customer_id</code>
3. Rename <code>customer_id;customer_card_2</code> to <code>customer_card</code>
4. Fill missing values with NaN in <code>customer_card</code>
5. Create column <code>Code_card</code> Extract value between positions 0 and 2 from <code>customer_card</code>
6. Create column <code>Code_card_1</code> using text function UPPER
7. Delete column <code>Code_card</code>
8. Create duplicate column <code>customer_id_copy</code> from <code>customer_id</code>
9. Change type of <code>customer_id_copy</code> to Number
10. Create column <code>RANDOM_BETWEEN</code> Column 1 using math function <code>RANDOM_BETWEEN</code>
11. Delete column <code>customer_id_copy</code>

Job (Trabajo): El proceso de ejecutar los pasos sobre el dataset se denomina trabajo.

Recipe jobs (3) Info					
Find jobs					Show all ▼
☐	Job name ▼	Status	Job input ▼		
☐	clientes-tarjetas	✓ Succeeded	cliente-tarjet... Project	(TipoDeTarjet... Dataset	+ cliente-tarjet... Recipe)
☐	order	✓ Succeeded	dbventas-order Project	(dbventas-sal... Dataset	+ dbventas-ord... Recipe)
☐	customer	✓ Succeeded	dbventas-clie... Project	(dbventas-sal... Dataset	+ dbventas-clie... Recipe)

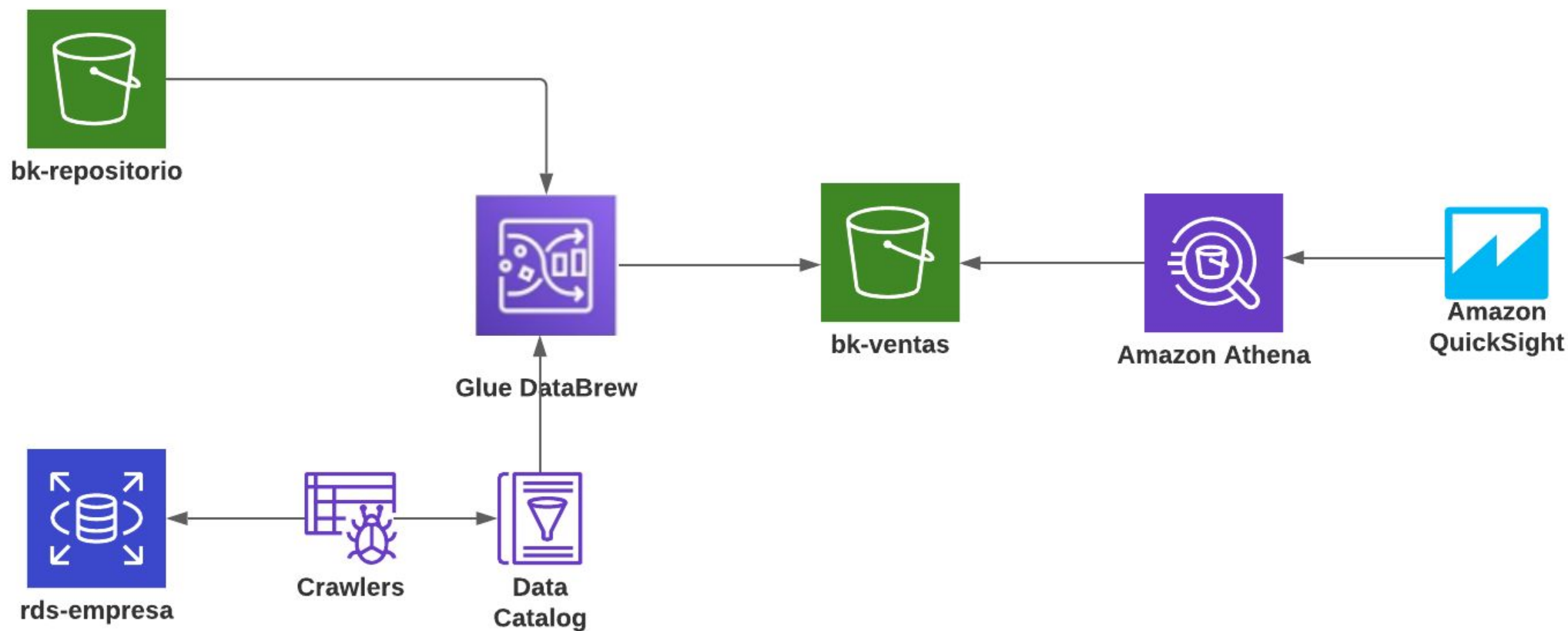
Arquitectura



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Fuente de Datos



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- Archivo CSV : id_customer y customer_card
- RDS MySQL: Esquema : Sales - customer

	customer_id	first_name	last_name	phone	email	street	city	state	zip_code
1	1	Debra	Burks	[NULL]	debra.burks@yahoo.com	9273 Thorne Ave.	Orchard Park	NY	14127
2	2	Kasha	Todd	[NULL]	kasha.todd@yahoo.com	910 Vine Street	Campbell	CA	95008
3	3	Tameka	Fisher	[NULL]	tameka.fisher@aol.com	769C Honey Creek St.	Redondo Beach	CA	90278
4	4	Daryl	Spence	[NULL]	daryl.spence@aol.com	988 Pearl Lane	Uniondale	NY	11553
5	5	Charolette	Rice	(916) 381-6003	charolette.rice@msn.com	107 River Dr.	Sacramento	CA	95820
6	6	Lyndsey	Bean	[NULL]	lyndsey.bean@hotmail.com	769 West Road	Fairport	NY	14450
7	7	Latasha	Hays	(716) 986-3359	latasha.hays@hotmail.com	7014 Manor Station Rd.	Buffalo	NY	14215
8	8	Jacqueline	Duncan	[NULL]	jacqueline.duncan@yahoo.com	15 Brown St.	Jackson Heights	NY	11372

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File upload

Data lake/data store

Amazon S3

AWS Glue Data Catalog

Amazon S3 tables

Amazon Redshift tables

Amazon RDS tables

All AWS Glue tables

Others

AWS Data Exchange

S3 pathParameterized S3 path

Enter your source from S3 [Info](#)
For you to select a folder, all files in the folder need to share the same file type. If there are different schemas, they will be merged.

s3://bk-repositorio/TipoDeTarjeta-Clientev1.csv

Format is: s3://bucket/prefix

TipoDeTarjeta-Clientev1.csv is selected

S3 Buckets > bk-repositorio

Select the entire folder

Search S3 objects by name

< 1 > ⚙

	Name	Size
☐	CientesTipoTarjeta	-
☐	miscanciones	-
☐	MisCanciones.csv	1.78 KB
☐	Nike Factory Locations.xlsx	93.91 KB
☐	TipoDeTarjeta-Cliente.csv	13.24 KB
☒	TipoDeTarjeta-Clientev1.csv	17.58 KB
☐	purchase-order-quantity-price-detail-for-commodity-goods-procurements-1.csv	89.01 MB

Dataset RDS MySQL



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- Crear una conexión hacia la instancia RDS en AWS GLUE
- AWS Glue no trabaja con MySQL v8
- Crear Crawler

Data stores	
Data store	JDBC
Connection	cn-rds-dbventas
Include path	production
Exclude paths	
Data store	JDBC
Connection	cn-rds-dbventas
Include path	sales
Exclude paths	

☐ production_brands	db-ventas
☐ production_categories	db-ventas
☐ production_products	db-ventas
☐ production_stocks	db-ventas
☐ sales_customers	db-ventas
☐ sales_order_items	db-ventas
☐ sales_orders	db-ventas
☐ sales_staffs	db-ventas
☐ sales_stores	db-ventas

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Amazon S3	Permission from AWS Lake Formation will apply to datasets with this icon.		
AWS Glue Data Catalog	AWS Glue databases > db-ventas		
Amazon S3 tables	Search tables by name		
Amazon Redshift tables	< 1 > ⚙		
Amazon RDS tables			
All AWS Glue tables			
Others			
AWS Data Exchange			
	☐	production_brands 12 hours ago January 24, 2021, 3:14:04 am	Not crawled
	☐	production_categories 12 hours ago January 24, 2021, 3:14:05 am	Not crawled
	☐	production_products 12 hours ago January 24, 2021, 3:14:05 am	Not crawled
	☒	production_stocks 12 hours ago January 24, 2021, 3:14:06 am	Not crawled
	☐	sales_customers 12 hours ago January 24, 2021, 3:14:06 am	Not crawled
	☐	sales_order_items 12 hours ago January 24, 2021, 3:14:07 am	Not crawled
	☐	sales_orders 12 hours ago January 24, 2021, 3:14:07 am	Not crawled
	☐	sales_staffs 12 hours ago January 24, 2021, 3:14:08 am	Not crawled
	☐	sales_stores 12 hours ago January 24, 2021, 3:14:08 am	Not crawled

Receta - Glue DataBrew



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- CVS tarjetas:
 - Crear la columna `id_card` usando un número aleatorio de tamaño 5 y uniendolo con el código de la tarjeta (CL-PR-DO)
- Unir la tabla **salescustomers** de otro Dataset

☒  **Inner join**
Select all rows that meet join condition from Table A and Table B.

☐  **Left join**
Select all rows from Table A and rows that meet join condition from Table B.

☐  **Right join**
Select all rows from Table B and rows that meet join condition from Table A.

☐  **Outer join**
Select all rows from Table A and Table B regardless of join condition.

☐  **Left excluding join**
Select all rows from Table A excluding the rows that meet join condition.

☐  **Right excluding join**
Select all rows from Table B excluding the rows that meet join condition.

Exportar Data



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Job output locations

Locations where the destination files are generated

File type

S3 location

CSV

 [s3://bk-ventas/](#) 

Objects (3)

Objects are the fundamental entities stored in Amazon S3. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#) 

 Find objects by prefix

☐	Name	▲	Type
☐	 clientes-tarjetas_24Jan2021_1611520052657/		Folder
☐	 customer_24Jan2021_1611480241481/		Folder
☐	 order_24Jan2021_1611482936321/		Folder

Presentar Datos



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- Crear Base de datos Ventas BI
- Crear Crawler para el nuevo archivo

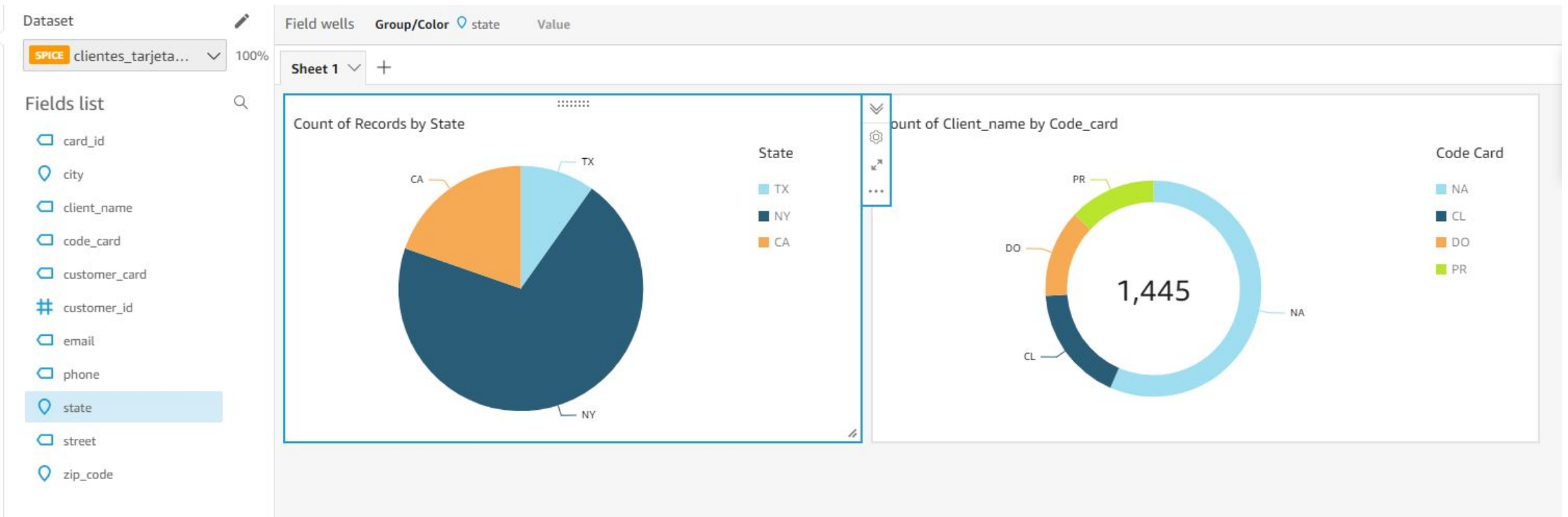
	Column name	Data type
1	customer_id	bigint
2	card_id	string
3	client_name	string
4	phone	string
5	customer_card	string
6	email	string
7	street	string
8	city	string
9	state	string
10	zip_code	bigint
11	code_card	string



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Preguntas



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#awsugperuconf



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gracias



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